

Ranked Retrieval 2: 概率模型、BM25 与语言模型

图文讲义版：用原 PPT 作为视觉锚点，按“概念故事线 + 直觉解释 + 考试速记”整理。

VSM heuristic → probabilistic uncertainty → PRP → relevance probability → BM25 → language model for IR → query likelihood → MLE zero problem → smoothing → model comparison

10/7/25

Recall: VSM & TFIDF term weighting

- Combines TF and IDF to find the weight of terms

$$w_{t,d} = (1 + \log_{10} tf(t, d)) \times \log_{10} \left(\frac{N}{df(t)} \right)$$

- For a query q and document d , retrieval score $f(q, d)$:

$$Score(q, d) = \sum_{t \in q \cap d} w_{t,d}$$

- TFIDF observations **Can we do better?**
 - Term appearing more in a doc gets higher weight (TF)
 - First occurrence is more important (log)
 - Rare terms are more important (IDF)
 - Bias towards longer documents

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IR Model

- VSM is very heuristic in nature
 - No notion of relevance is there (still works well)
 - Any weighting scheme, similarity measure can be used
 - Components not interpretable → no guide for what to try next
 - More engineering rather than theory → tweak, run, observe, tweak ...
 - Very popular, hard to beat, strong baseline
 - Easy to adapt good ideas from other models
- Probabilistic Model** of retrieval
 - Mathematical formulation for relevant / irrelevant sets
 - Explicitly defines random variables (R, Q, D)
 - Specific about what their values are
 - State the assumptions behind each step
 - Watch out for contradictions

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这节课一句话

这节课把 ranking 从 TF-IDF 的工程配方推进到概率视角：文档为什么相关、BM25 为什么稳、语言模型为什么能用“生成 query”的概率排序。

1. TF-IDF 很强，但理论感偏弱

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人话直觉

TF-IDF 像一套经验丰富的厨师配方：很好吃，但你还想知道每个调料为什么该这么放。

精确定义 / 机制： VSM/TF-IDF 没有显式建模 relevance，只是组合 weighting scheme 和 similarity measure。它非常流行、强 baseline、难以击败，但 components 不完全可解释，也容易依赖 tweak-run-observe 的工程循环。

考试问法： 若问为什么引入 probabilistic model，答 VSM heuristic、缺少 explicit relevance notion、概率模型能定义 random variables 和 assumptions。

2. PRP: 按相关概率排序

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Formulation of PRP

- Rank docs by probability of relevance
 - $P(R|D_{r1}) > P(R|D_{r2}) > P(R|D_{r3}) > P(R|D_{r4}) > \dots$
- Estimate probability as accurate as possible
 - $P_{\text{est}}(R|D) \approx P_{\text{true}}(R|D)$
- Estimate with all possibly available data
 - $P_{\text{est}}(R | \text{doc, session, context, user profile, ...})$
- Best possible accuracy can be achieved with that data
 - the perfect IR system
 - Is it really doable?
- How to estimate the probability of relevance?

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PRP Concept

- Imagine IR as a classification problem

Document

Relevant Documents

Non-Relevant Documents

$P(R|D) + P(NR|D) = 1$

- Document D is relevant if $P(R|D) > P(NR|D)$

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人话直觉

如果你能估计每篇文档“正中用户意图”的概率，那么最合理的事就是从最高概率开始排。

精确定义 / 机制： Probability Ranking Principle 认为，在可用数据下，如果系统能尽可能准确估计每个 document 对请求的 relevance probability，那么按该概率递减排序能获得最佳 effectiveness。

$$P(R = 1 | d_1, q) > P(R = 1 | d_2, q) \Rightarrow d_1 > d_2$$

考试问法： PRP 答题核心：rank by decreasing probability of relevance, probabilities estimated as accurately as possible from available data。

3. 相关概率不是读心术

10/7/25

Probability of Relevance

- What is $P_{\text{true}}(\text{rel} \mid \text{doc, query, session, user, ...})$?
 - Isn't relevance just the user's opinion?
 - User decides relevant or not, what is the "probability" thing?
- Search algorithm cannot look into your head (yet!)
 - Relevance depends on factors that algorithm cannot observe
 - SIGIR 2016 best paper award: *Understanding Information Need: an fMRI Study*
- Different users may disagree on relevance of the same doc
 - Even similar users, doing the same task, in the same context
- $P_{\text{true}}(\text{rel} \mid Q, D)$:
 - Proportion of all unseen users / context / tasks for which D would have judged relevant to Q
- Similar to: $P(\text{die}=6 \mid \text{even and not square})$

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Okapi BM25 Model

- Based on the probabilistic model
 - A document D is relevant if $P(R=1|D) > P(R=0|D)$
- Extension to the "binary independence model"
 - Binary features:** Document represented by a vector of binary features indicating term occurrence
 - Assume **term independence** (Naïve Bayes assumption)
 - BOW trick
- In 1995, *Stephan Robertson* with his group came up with the **BM25** Formula as part of the **Okapi** project.
- It outperformed all other systems in TREC
- Popular and effective ranking algorithm

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人话直觉

系统看不到你脑子里的真实 need，只能根据 query、document、context、user profile 等影子去猜。

精确定义 / 机制：真实 relevance 受用户、任务、上下文影响，不同用户可能对同一 document 判断不同。 $P_{\text{true}}(\text{rel} \mid Q, D)$ 可理解为在未观察用户/情境/任务中，D 会被判 relevant 的比例；系统只能估计 P_{est} 。考试问法：若问 relevance probability 如何理解，不要说用户主观就没概率；要说的对不可见用户/场景的比例性估计。

4. BM25: 概率模型落地成强公式

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Okapi BM25 Ranking Function

- Let L_d be the number of terms in document d
- Let $\bar{L}$ be the average number of terms in a document

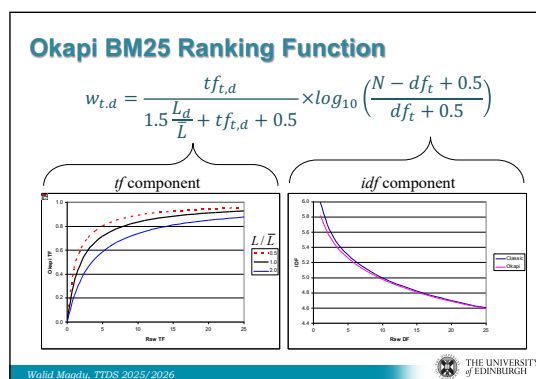
$$w_{t,d} = \frac{tf_{t,d}}{k \cdot \frac{L_d}{\bar{L}} + tf_{t,d} + 0.5} \times \log_{10} \left(\frac{N - df_t + 0.5}{df_t + 0.5} \right)$$

- Best practices: $k=1.5$

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人话直觉

BM25 像 TF-IDF 的升级款避震车：仍奖励关键词出现和稀有性，但更稳地处理重复词和长文档。

精确定义 / 机制： BM25 源于 probabilistic retrieval/Binary Independence Model 的扩展，是经典且强大的 ranking function。它包含 term frequency saturation、IDF 和 document length normalization； k 控制 tf saturation，文档长度用 L_d/L_{avg} 调整。

$$w_{t,d} = \frac{tf_{t,d}}{tf_{t,d} + k \left((1 - b) + b \frac{L_d}{L_{avg}} \right)} \log_{10} \frac{N - df_t + 0.5}{df_t + 0.5}$$

考试问法： 若考 BM25，答 saturated TF、IDF、length normalization，并说明 k 控制 tf 增长饱和。

5. BM25 为什么比朴素 TF-IDF 稳

10/7/25

Probabilistic Model in IR

- Focuses on the probability of relevance of docs
- Could be mathematically proved
- Different ways to apply it
- BM25 is the most common formula for it

- What other models could be still used in IR?

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"Noisy-Channel" Model of IR

Information need

User has a information need and writes down a query

Query

Machine's task: Given the query, guess which document matches the query.

document collection

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人话直觉

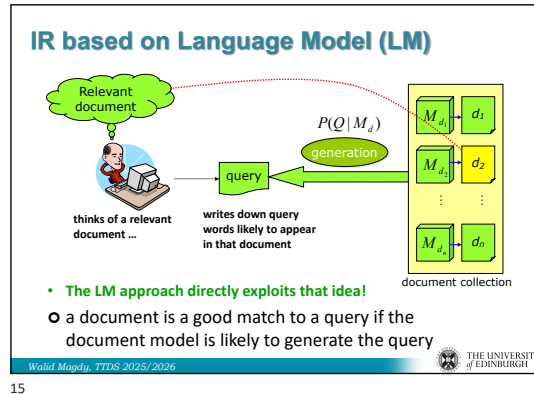
一个词出现 2 次比 1 次重要，但出现 200 次不该比 100 次再重要一倍；BM25 会让这种增长慢慢饱和。

精确定义 / 机制： BM25 与 TF-IDF 都奖励 document 内 query term 出现和 collection rare terms，但 BM25 更明确地处理 tf saturation 和 document length bias。长文档不会仅因词多而持续占优，重复词也不会无限加分。

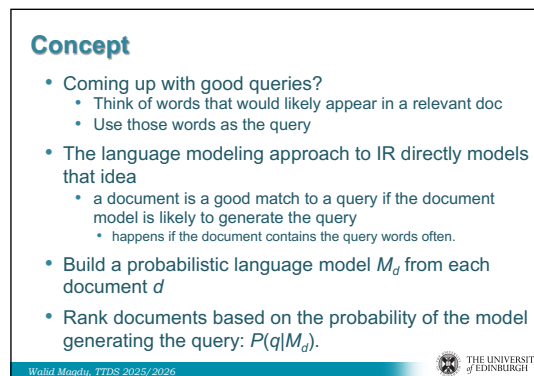
考试问法： 比较 BM25 vs TF-IDF 时，重点说 BM25 显式加入 saturation 与 length normalization。

6. Language Model for IR: 好文档会“说出” query

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人话直觉

如果一篇文档的语言习惯很容易吐出你的 query，那么它很可能在讲你要找的东西。

精确定义 / 机制：LM approach 为每个 document 建 language model，并按 query likelihood $P(q|M_d)$ 排序。Bayes rule 下 $P(d|q)$ 与 $P(q|d)$ 相关，若 $P(q)$ 对所有 documents 相同、 $P(d)$ 作为相同 prior 或另行处理，就可以比较 $P(q|M_d)$ 。

$$P(d | q) = \frac{P(q | d)P(d)}{P(q)}$$

考试问法：若问 LM for IR 核心思想，答 document language model 生成 query 的概率越高，document 越相关。

7. Unigram LM 与 query likelihood

10/7/25

$P(q|d)$ Query Likelihood Model

- We will make the conditional independence assumption.

$$P(q|M_d) = P(\langle t_1, \dots, t_{|q|} \rangle | M_d) = \prod_{1 \leq k \leq |q|} P(t_k | M_d)$$

$|q|$: length of q ; t_k : token occurring at position k in q

- This is equivalent to:

$$P(q|M_d) = \prod_{\text{each term } t \text{ in } q} P(t|M_d)^{tf_{t,q}}$$

$tf_{t,q}$: term frequency (# occurrences) of t in q

- Multinomial model (omitting constant factor)

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Parameter estimation

- Probability of a term t in a LM M_d using Maximum Likelihood Estimation (MLE)

$$P(t|M_d) = \frac{tf_{t,d}}{|d|}$$

$|d|$: length of d ;
 $tf_{t,d}$: # occurrences of t in d

- Probability of a query q to be noticed in a LM M_d :

$$P(q|M_d) = \prod_{t \in q} \left(\frac{tf_{t,d}}{|d|} \right)^{tf_{t,q}}$$

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人话直觉

先假装 query 里的词像从一个词袋里独立抽球，这样复杂句子概率就能拆成单词概率乘积。

精确定义 / 机制： unigram LM 假设 terms 条件独立，不考虑顺序。query likelihood model 将 $P(q|M_d)$ 分解为 query terms 在 document model 下概率的乘积，也可按 query term frequency 写成幂次形式。

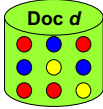
$$P(q | M_d) = \prod_{t \in q} P(t | M_d)^{f(t,q)}$$

考试问法： 公式题要写 conditional independence assumption，并说明 M_d 是 document language model。

8. MLE 的零概率问题

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Example

$$P(\bullet \bullet \bullet \bullet) = P(\bullet)^2 \times P(\bullet) \times P(\bullet) = (4/9)^2 \times (2/9) \times (3/9) = 0.0146$$
$$P(\bullet \bullet \bullet \bullet)$$


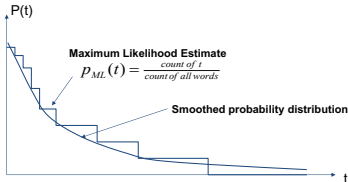
- Is that fair?
 - In VSM, $S(Q,D)$ was summation, works more like OR in Boolean search. Missing one term reduces score only
 - In language model, $S(Q,D)$ is $P(Q|D) \rightarrow$ Multiplication of probabilities $\rightarrow$ missing one term makes score = 0
 - Is there a better way to handle unseen terms?

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Smoothing

- Problem: Zero frequency
- Solution: "Smooth" terms probability



Maximum Likelihood Estimate
 $P_{ML}(t) = \frac{\text{count of } t}{\text{count of all words}}$

Smoothed probability distribution

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人话直觉

一篇文章没出现某个词，不代表它的主题模型绝不可能说这个词；样本没见到不等于世界不存在。

精确定义 / 机制：MLE 通常用 $tf(t,d)/|d|$ 估计 document LM。若 query term 没出现在 document 中，概率为 0，整个 query likelihood 乘积变 0。这对检索过于严厉，因为 document text 只是语言模型的一个 sample。

$$P_{MLE}(t | M_d) = \frac{tf(t,d)}{|d|}$$

考试问法：若问为什么需要 smoothing，答 zero frequency/data sparsity: unseen query term 会让 $P(q|M_d) = 0$ 。

9. Smoothing: 给没见过的词留一口气

10/7/25

Smoothing

- Document texts are a sample from the language model
- Missing words should not have zero probability of occurring
- A missing term is possible (even though it didn't occur)
 - but no more likely than would be expected by chance in the collection.
- A technique for estimating probabilities for missing (or unseen) words
 - Overcomes data-sparsity problem
 - lower (or discount) the probability estimates for words that are seen in the document text
 - assign that "left-over" probability to the estimates for the words that are not seen in the text (and also on the seen ones)

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Mixture Model

$$P(t|d) = \lambda P(t|M_d) + (1 - \lambda)P(t|M_c)$$

- Mixes the probability from the document with the general collection frequency of the word.
- Estimate for unseen words is $(1-\lambda) P(t|M_c)$
 - Based on collection language model (background LM)
 - $P(t|M_c)$ is the probability for query word t in the collection language model for collection C (background probability)
 - λ is a parameter controlling probability for unseen words
- Estimate for observed words is
$$\lambda P(t|M_d) + (1-\lambda) P(t|M_c)$$

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人话直觉

没在这篇文档里见过的词，不该判死刑；可以向整个 collection 借一点背景概率。

精确定义 / 机制：smoothing 给 unseen terms 分配非零概率，同时降低 observed terms 的过度自信估计。mixture/Jelinek-Mercer smoothing 将 document LM 与 collection LM 线性混合，lambda 控制文档证据和背景语言的权衡。

$$P(t | d) = \lambda P(t | M_d) + (1 - \lambda)P(t | M_c)$$

考试问法：若考 smoothing，说明目标是解决 zero probability，并写 document LM 与 collection LM 的混合。

10. λ 的检索行为: conjunctive vs disjunctive

10/7/25

Jelinek-Mercer Smoothing

$$P(t|d) = \lambda P(t|M_d) + (1 - \lambda)P(t|M_c)$$

- **High value of λ :** "conjunctive-like" search – tends to retrieve documents containing all query words.
- **Low value of λ :** more disjunctive, suitable for long queries
- Correctly setting λ is important for good performance.
- Final Ranking function:
$$P(q|M_d) \propto \prod_{1 \leq k \leq |q|} (\lambda \cdot P(t_k|M_d) + (1 - \lambda) \cdot P(t_k|M_c))$$

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Example

- **Collection:** d_1 and d_2
- d_1 : "Jackson was one of the most talented entertainers of all time"
- d_2 : "Michael Jackson anointed himself King of Pop"
- **Query q :** Michael Jackson
- Use mixture model with $\lambda = 1/2$
- $P(q|d_1) = [(0/11 + 1/18)/2] \cdot [(1/11 + 2/18)/2] \approx 0.003$
- $P(q|d_2) = [(1/7 + 1/18)/2] \cdot [(1/7 + 2/18)/2] \approx 0.013$
- Ranking: $d_2 > d_1$

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人话直觉

λ 高时更执着于本文档有没有这些词，像 AND； λ 低时更宽容，像 OR 风格的长 query 检索。

精确定义 / 机制：Jelinek-Mercer 中高 lambda 更依赖 document LM，倾向检索包含所有 query words 的 documents，表现为 conjunctive-like。低 lambda 更多借 collection background，行为更 disjunctive，适合 long queries。

考试问法：lambda 行为常考：高 λ 更 conjunctive-like；低 λ 更 disjunctive/long-query friendly。

11. 三种模型问的是不同问题

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LMs for IR: 3 possibilities

- Probability of generating the query text from a document language model
- Probability of generating the document text from a query language model
- Comparing the language models representing the query and document topics

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Summary

- **Three ways to model IR**
- VSM
How query vector aligns with document vector?
- Probabilistic Model
What is the relevance probability of document D given query Q?
- LM
How likely is it possible to observe/generate sequence of terms Q in a language model of document D?

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人话直觉

VSM 问“向量像不像”，概率模型问“相关概率多大”，LM 问“这篇文档能不能生成这个 query”。

精确定义 / 机制：VSM 使用 weighting 与 similarity；probabilistic model 显式建模 relevance probability；LM for IR 则用 language model 的生成概率解释匹配。三者都可有效，BM25 和 query likelihood LM 往往效果相近，更复杂方法可能超过 BM25。

考试问法：模型比较题要按核心问题区分：alignment、relevance probability、query generation likelihood。

考试速记版

- VSM/TF-IDF 很强但 heuristic，没有显式 relevance probability。
- PRP：按 decreasing probability of relevance 排序，估计越准越好。

- $P_{true}(rel|Q, D)$ 可理解为跨未观察用户/情境的相关比例。
- BM25 = saturated TF + IDF + document length normalization。
- LM for IR: document language model 生成 query 的概率越高, 排名越靠前。
- MLE 会让 unseen query term 概率为 0, 因此需要 smoothing。
- Jelinek-Mercer: $\lambda P(t|M_d) + (1-\lambda)P(t|M_c)$, λ 控制 conjunctive/disjunctive 行为。

一段稳妥考场回答

Probabilistic retrieval 的核心是把 IR 中的不确定性形式化, 并按 relevance probability 排序。PRP 认为, 如果能基于可用数据尽可能准确估计 $P(R=1|d,q)$, 按该概率递减排序可获得最优 effectiveness。BM25 是概率检索思想中最重要的实际公式之一, 它结合 saturated term frequency、IDF 和 document length normalization, 因此比朴素 TF-IDF 更稳。Language Model for IR 则为每个 document 建 LM, 并按 $P(q|M_d)$ 即 document model 生成 query 的概率排序; MLE 会导致 unseen terms 概率为 0, 所以需要 smoothing, 如 Jelinek-Mercer 用 $\lambda P(t|M_d) + (1-\lambda)P(t|M_c)$ 混合文档模型和 collection 背景模型。